CONTROL NARRATIVE DOCUMENT

Pump Station PS-01 — Rev B (Working Draft)

Document Number: CND-PS01-002

Revision: B (DRAFT — NOT APPROVED)

Date: 29 October 2026

1. GENERAL DESCRIPTION

Pump Station PS-01 transfers process fluid from suction vessel T-100 to the downstream process unit. Two centrifugal pumps operate in duty/standby configuration with automatic changeover. Flow is regulated by FV-301 based on FIC-201 setpoint.

1.1 P&ID Schematic

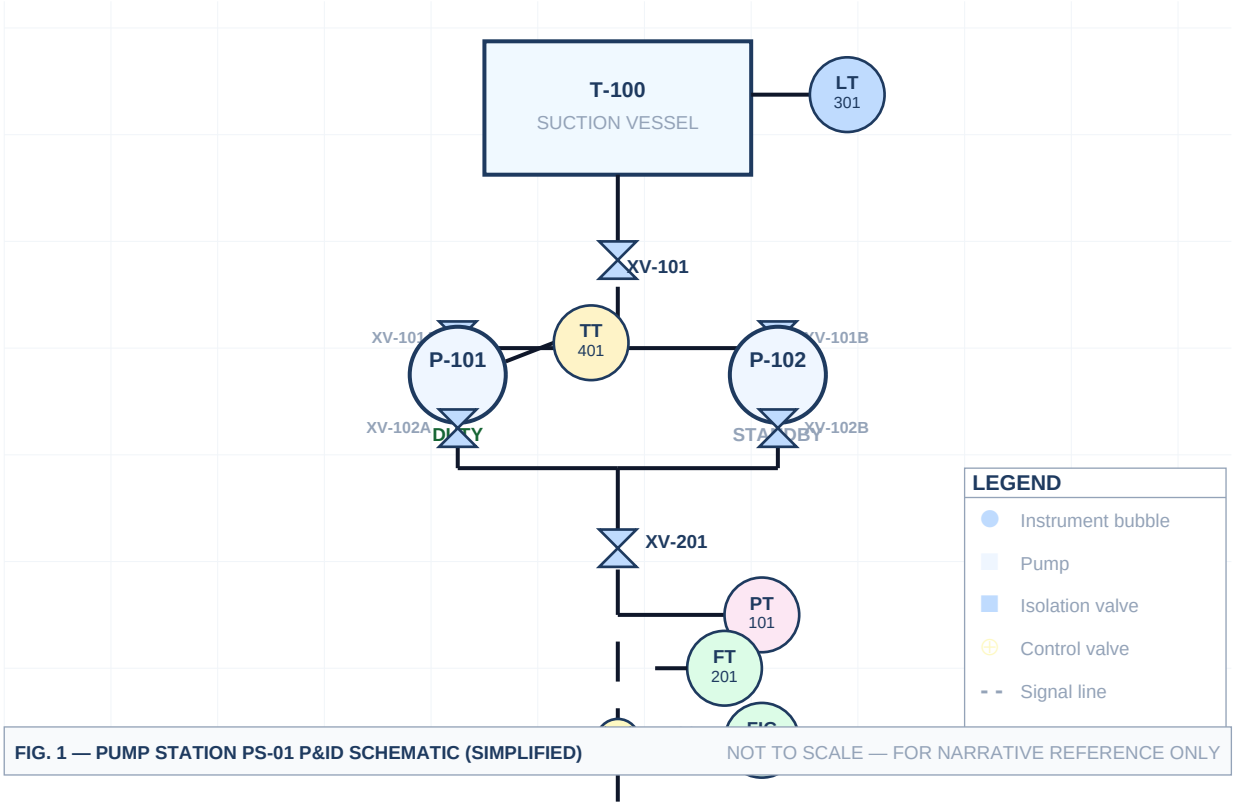

FIG. 1 — PUMP STATION PS-01 P&ID SCHEMATIC (SIMPLIFIED) NOT TO SCALE — FOR NARRATIVE REFERENCE ONLY

2. EQUIPMENT LIST

Tag	Description	Type	Range / Rating	Normal State
P-101	Duty pump	Centrifugal pump	0–150 m³/h, 11 kW	Running
P-102	Standby pump	Centrifugal pump	0–150 m³/h, 11 kW	Stopped
XV-101	Suction isolation valve	Pneumatic gate valve	DN150, PN16	Open
XV-201	Main discharge valve	Pneumatic gate valve	DN200, PN16	Open
FV-301	Flow control valve	Pneumatic globe valve	Cv 120, fail-open	Auto
PT-101	Discharge pressure	Pressure transmitter	0–300 PSI	Active
FT-201	Process flow	Magnetic flow transmitter	0–200 L/min	Active
LT-301	Suction vessel level	Radar level transmitter	0–5 m	Active
TT-401	Pump bearing temperature	RTD temperature transmitter	0–150 °C	Active

3. OPERATING MODES

3.1 STOPPED

Both pumps de-energised. XV-201 closed. Entry: operator or fault.

3.2 STARTING

Startup sequence executing. Duration up to 120 seconds.

3.3 RUNNING

P-101 running. FIC-201 controlling FV-301 to 80 L/min setpoint.

3.4 STANDBY

P-101 stopped, P-102 auto-started. Alarm generated.

3.5 STOPPING

Shutdown sequence executing. Duration up to 60 seconds.

4. SETPOINTS TABLE

Tag	Parameter	Value	Units	Page Ref
PT-101	High High Trip	150	PSI	p.3
PT-101	High Alarm	130	PSI	p.3
FT-201	Low Flow Trip	10	L/min	p.3
FT-201	Low Alarm	20	L/min	p.3
LT-301	Low Low Trip	15	%	p.3
TT-401	High High Trip	85	°C	p.3

5. INTERLOCKS

Interlock IL-101 — High High Discharge Pressure Trip

Field	Content
Cause	PT-101 discharge pressure exceeds 150 PSI for more than 3 seconds
Action	Stop P-101 AND stop P-102 AND close XV-201 AND alarm on HMI
Reset Condition	PT-101 falls below 120 PSI AND operator investigated root cause
Reset Method	Manual reset — operator presses RESET on HMI Panel
Priority	Safety Critical

Interlock IL-201 — Low Low Flow Trip

Field	Content
Cause	FT-201 flow rate falls below 10 L/min for more than 5 seconds while pump running
Action	Stop pump P-101 AND auto-start standby pump P-102 AND alarm 'LOW FLOW TRIP'

Interlock IL-301 — Low Low Suction Level Trip

Field	Content
Cause	LT-301 suction vessel level falls below 15% for more than 10 seconds
Reset Condition	LT-301 level above 25% AND operator confirms feed restored
Reset Method	Manual reset — operator presses RESET on HMI Panel
Priority	Safety Critical

6. ALARMS

Tag	Type	Setpoint	Units	Priority	Operator Response
PT-101-HH	HH	150	PSI	Priority 1	Stop all pumps. Close XV-201. Investigate.
PT-101-H	H	130	PSI		Reduce FIC-201 setpoint. Monitor PT-101.
FT-201-L	L	20	L/min	Priority 2	Check FV-301 position. Verify P-101 running.
LT-301-L	L	25	%	Priority 2	Contact feed ops. Prepare for pump shutdown.
TT-401-H	H	75	°C	Priority 2	Reduce load or switch to P-102.